Hints: Functional sequences, series and spaces – 2. Series of functions

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. All three tests are statements about the tails. Weierstrass: $\left|\sum_{n=N}^{M} u_n(x)\right| \leqslant \sum_{n=N}^{M} c_n$ and the uniform Cauchy criterion. Dirichlet and Abel: apply Abel summation $\sum_{n=N}^{M} u_n v_n = u_M V_M - u_N V_{N-1} - \sum_{n=N}^{M-1}(u_{n+1}-u_n)V_n$ (with V_n the partial sums of v, or the tails in Abel's case) and estimate everything uniformly in x; monotonicity makes $\sum|u_{n+1}-u_n|$ telescope.

Hint 2. $\sum \frac{\sin nx}{n^\alpha}$: for $\alpha > 1$ uniform on $\mathbb{R}$ (Weierstrass). For $0 < \alpha \leqslant 1$ use $\left|\sum_{n=1}^{N} \sin nx\right| \leqslant \frac{1}{|\sin(x/2)|}$: Dirichlet gives uniform convergence on every $[\delta, 2\pi-\delta]$ (and its translates by $2\pi k$), convergence everywhere. Near 0 it is not uniform: at $x = \frac{1}{2N}$ the block $\sum_{n=N}^{2N}$ has all terms at least $\frac{\sin(1/2)}{(2N)^\alpha}$, so it is at least $c\,N^{1-\alpha} \geqslant c$.

Hint 3. $\sum \frac{(-1)^n}{n+x^2}$: the Leibniz tail estimate $\leqslant \frac{1}{N+1+x^2} \leqslant \frac{1}{N+1}$ gives uniform convergence on all of $\mathbb{R}$, while the convergence is absolute nowhere. $\sum x^n(1-x)$ converges on $(-1,1]$ (a geometric series; the terms vanish at $x=1$); its tail from N is x^N for $|x|<1$, so the convergence is uniform on $[-a,b]$ for $0<a,b<1$ but not on $[0,1)$, $[0,1]$ or near -1; on $[0,1]$ the sum is discontinuous at 1.

Problem 2

Hint 1. For smoothness, bound all derivatives of $\frac{1}{n^2+x^2}$ uniformly in x. A clean way: $\frac{1}{n^2+x^2} = \frac{1}{2in}\left(\frac{1}{x-in} - \frac{1}{x+in}\right)$, so the k-th derivative is at most $\frac{k!}{n^{k+2}}$ in absolute value; apply the M-test to every differentiated series and Theorem 3.

Hint 2. Asymptotics: $t \mapsto \frac{1}{t^2+x^2}$ is decreasing, so $\int_1^\infty \frac{dt}{t^2+x^2} \leqslant S(x) \leqslant \int_0^\infty \frac{dt}{t^2+x^2} = \frac{\pi}{2x}$; hence $S(x) \sim \frac{\pi}{2x}$ as $x \to \infty$ (in fact $S(x) = \frac{\pi x \coth \pi x - 1}{2x^2}$, if you know it).

Hint 3. $\sum \frac{\sin nx}{n}$ converges everywhere by Dirichlet. Its sum is $\frac{\pi - x}{2}$ on $(0, 2\pi)$ (integrate $\frac{1}{2} + \sum_{n=1}^{N} \cos nt = \frac{\sin(N+\frac{1}{2})t}{2\sin(t/2)}$ from x to π and let $N \to \infty$ using the Riemann–Lebesgue lemma), and 0 at $x = 0$. The jump at 0 shows the convergence cannot be uniform on any neighbourhood of 0: a uniform limit of continuous functions is continuous.

Problem 3

Hint 1. Uniform convergence is the M-test with $c_n = 1/n^2$. Each term $\{nx\}/n^2$ is right-continuous and is continuous except at the points with $nx \in \mathbb{Z}$, where it drops by $1/n^2$.

Hint 2. Because the convergence is uniform, one-sided limits can be taken term by term. At an irrational x every term is continuous, so R is continuous. At $x = p/q$ in lowest terms, the terms that jump are those with $q \mid n$, and $R(x^-) - R(x) = \sum_{q|n} \frac{1}{n^2} = \frac{\pi^2}{6q^2} > 0$.

Hint 3. Integrability: each partial sum is a bounded function with finitely many discontinuities, hence integrable, and a uniform limit of integrable functions is integrable. (Alternatively, use Lebesgue's criterion: the discontinuity set is countable.)

Problem 4

Hint 1. Guess the limit: a fixed point satisfies $f' = \frac{1}{1+f}$, $f(0) = 0$, i.e. $\big((1+f)^2\big)' = 2$, so $f(x) = \sqrt{1+2x} - 1$.

Hint 2. Write $Tg(x) = \int_0^x \frac{dt}{1+g(t)}$. For $g, h \geqslant 0$, $|Tg - Th|(x) \leqslant \int_0^x |g-h|(t)\,dt$, because $\left|\frac{1}{1+g} - \frac{1}{1+h}\right| = \frac{|g-h|}{(1+g)(1+h)} \leqslant |g-h|$. All f_n are $\geqslant 0$.

Hint 3. Iterate this Volterra-type estimate against the fixed point $f = Tf$: $|f_n - f|(x) \leqslant \frac{x^n}{n!} \max|f_0 - f|$. So $f_n \to \sqrt{1+2x} - 1$, and the convergence is uniform on $[0,1]$.

Problem 5

Hint 1. Each f_n is a polynomial with $f_n' = f_{n-1}$ and $f_n(a_n) = 0$. Compute the first cases: $f_1(x) = x - a_1$, and $f_2(0) = 0$ forces $a_2(a_2 - 2a_1) = 0$.

Hint 2. Since $f_n' = f_{n-1}$, every f_n is $\frac{x^n}{n!} + c_1\frac{x^{n-1}}{(n-1)!} + \cdots + c_n$ with $c_k = f_k(0)$ (a generating function $\sum f_n(x)s^n = C(s)e^{sx}$ makes this transparent). The condition says $c_k = 0$ for $k \geqslant 2$, so by induction $f_n(x) = \frac{x^n}{n!} - a_1\frac{x^{n-1}}{(n-1)!} = \frac{x^{n-1}(x-na_1)}{n!}$.

Hint 3. In the induction step, f_n is that polynomial plus the constant that makes it vanish at a_n; the constant is zero iff $a_n^{n-1}(a_n - na_1) = 0$. Answer: a_1 arbitrary, and for each $n \geqslant 2$ independently $a_n = 0$ or $a_n = na_1$.